Inventory Optimisation & Working Capital Analysis

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1. Executive Summary

An end-to-end Python-based analysis was conducted on a sample warehouse dataset of 150 SKUs to evaluate current inventory health. The objective was to identify excess working capital tied up in overstocked items and flag critical stockout risks. By implementing a statistical ABC-XYZ classification and recalculating optimal safety stock levels, the analysis identified **[Insert your total $ amount here]** in trapped working capital that can be released back to the business, alongside several immediate fulfillment risks.

2. Methodology

The analysis was performed using Python (Pandas/NumPy) to move away from static spreadsheet calculations toward dynamic inventory parameters:

- **ABC Classification:** Categorized SKUs by annual financial value (80/15/5 Pareto principle).
- **XYZ Classification:** Measured demand volatility using the Coefficient of Variation (CV) to separate predictable items from erratic items.
- **Safety Stock & Reorder Points:** Calculated dynamic safety stock using the statistical formula ($SS = Z \times \sigma_{d} \times \sqrt{L}$), assigning higher service levels (Z-scores) to 'A' class items to ensure high availability without over-investing in 'C' class buffer stock.

3. Key Findings (Examples)

The script compared actual stock on hand against newly calculated maximum healthy stock levels, revealing two distinct operational issues:

- **Significant Trapped Capital (Overstock):** Several SKUs are severely over-ordered. For example, **SKU_070** currently holds 2,096 units. The calculated healthy maximum is only 1,693 units. This single item is currently trapping **$198,884.53** in excess capital. Similarly, **SKU_013** is tying up over $20,000 in excess buffer stock.
- **Critical Stockout Risks (Understock):** Conversely, procurement parameters for other items are failing to keep up with lead time demand. **SKU_089** is a critical risk; it currently holds 913 units on hand, but the statistical Reorder Point to prevent a stockout during supplier lead time is 1,643 units. The business is highly likely to fail on upcoming customer orders for this item.

4. Recommended Actions

1. **Halt Procurement on Overstocked Items:** Immediately freeze purchase orders for SKU_070 and SKU_013 until cycle stock burns down below the newly established Reorder Points.

2. **Expedite At-Risk Inventory:** Raise urgent purchase orders or expedite shipping for SKU_089 and other items operating below their emergency Reorder Points to protect customer fulfillment.
3. **Parameter Update (ERP/SAP):** Update the master data in the ERP system with the newly calculated Reorder Points (ROP) and Maximum Stock levels so future automated purchasing aligns with actual demand volatility, not historical guesswork.